BIOPULSE EVALUATION METRICS

Overview

This document outlines the evaluation methodology used to assess the BioPulse high-fidelity prototype. The evaluation was designed to validate the emergency-response workflows proposed in Milestone 1 and assess the usability, learnability, and effectiveness of the prototype.

The detailed evaluation results and discussion are presented in the final report

Experimental Setup & Evaluation Methodology

To evaluate the **BioPulse** mobile prototype against the quality criteria established in Milestone 1 (ISO/IEC 25010 and PACMAD frameworks), an empirical usability study was designed and executed.

Experimental Setup

- **Testing Environment:** Quiet usability testing laboratory setting to establish baseline task performance without initial ambient distractions.
- **Hardware & Software Setup:** The high-fidelity interactive prototype was deployed on an **iPhone 17** and a **Samsung Galaxy S21** via the Figma Mobile Engine, ensuring physical touchscreen ergonomics (hand-held reachability and high-stress thumb zones).
- **Hardware Simulation Protocol:** Because the **SkinGrip wearable patch has not yet been physically manufactured** for this design assignment, hardware interactions were evaluated using software-simulated triggers:
    - **NFC Pairing Simulation:** Users brought an NFC test card to the back of the phone to trigger the app's automated pairing animation sequence.
    - **Physiological Threat Simulation:** An automated script injected a simulated **GSR micro-sweat spike** into the prototype stream after 30 seconds of passive tracking to trigger the emergency alert countdown.
- **Participant Cohort ($N = 3$):** Three target users were recruited, each representing one of the project's core persona profiles:
    1. **User 1:** Matching *Sarah Williams* (Vulnerable Commuter profile — high safety anxiety, low setup patience).
    2. **User 2:** Matching *Michael Chen* (Independent Traveller profile — tech-savvy, values low-maintenance tracking).
    3. **User 3:** Matching *Margaret Taylor* (Independent Senior profile — lower tech literacy, requires high visual accessibility).

Measurement Protocol

1. **Quantitative Telemetry:** Screen recording software with millisecond touch-timestamp tracking recorded three artefacts: *Time-on-Task (ToT), Touch Count (Taps), and Unintended Error Actions.*
2. **Qualitative Metrics:** Post-task System Usability Scale (SUS) 10-item questionnaires and concurrent Think-Aloud Protocol notes.

Usability Testing Tasks

Instead of basic alarm creation, tasks focused strictly on high-stress safety and emergency escalation workflows:

Task 1: Simulated Wearable Onboarding & NFC Coupling Protocol

- *Procedure:* The participant is instructed to open the application, navigate to the device configuration interface, and simulate physical contact between an NFC test card and the smartphone's upper antenna region to establish the wearable sensor handshake.
- *Objective Measurement:* Verify whether the user can initialize device coupling in a single interaction step without encountering navigational ambiguity or configuration errors.

Task 2: Passive Operational State & Status Telemetry Assessment

- *Procedure:* The participant is presented with the primary monitoring dashboard during an active monitoring session and instructed to visually identify and verbally state three operational parameters within a 3-second window: sensor connection integrity, wearable battery percentage, and real-time GSR signal activity.
- *Objective Measurement:* Assess visual hierarchy clarity, visual salience, and rapid situation awareness under normal passive operation.

Task 3: Rapid Biometric False-Alarm Disarm Execution

- *Procedure:* Upon the automated triggering of an acute GSR threat alert and 5-second countdown timer, the participant is instructed to intervene immediately by locating and executing the primary cancellation control using native Face ID / Touch ID biometric authentication.
- *Objective Measurement:* Measure execution time, cognitive load, and touch accuracy during high-salience threat alert states to confirm false alarms can be disarmed within the safety window.

Task 4: Emergency Escalation & Automated Dispatch Verification

- *Procedure:* The participant is presented with an active threat alert and instructed to withhold intervention, allowing the 5-second countdown timer to

elapse completely. The participant must then observe the resulting interface state to verify the automatic assembly and transmission of emergency telemetry (live GPS coordinates and contact notifications).

- *Objective Measurement:* Validate end-to-end automated escalation behaviour and confirm clear post-dispatch visual feedback and status reassurance.